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Consider the following processes with their arrival times and burst times:

Using Shortest-Job-First (SJF) scheduling (non-preemptive), what is the waiting time for process P4?

Process	Arrival Time	Burst Time	Priority
P1	0	8	3
P2	1	4	1
P3	2	9	4
P4	3	5	2



13

9

11

6

Will be reviewed

9

What is "bounded waiting" in the context of the critical-section problem?

Limiting the number of processes that can access the critical section.

Limiting the amount of memory used by a process.

Ensuring that a process waits for a limited time to enter the critical section.

Preventing processes from entering the critical section.

Will be reviewed

10

What is the critical section? 

A section of the disk used for storing system files.

A section of the kernel that handles interrupts.

A section of code that accesses shared resources.

A section of memory used for storing critical data.

 **Will be reviewed**

11

What is the purpose of an interrupt vector table? 

To store the file system structure.

To store the addresses of interrupt handlers.

To store the system's memory map.

To store the list of running processes.

 **Will be reviewed**

12

Consider the following processes with their arrival times and burst times:
Using First-Come, First-Served (FCFS) scheduling, what is the waiting time for process P3?



21

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19

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Will be reviewed

13

Consider the following processes with their arrival times and burst times:
Using Priority scheduling (non-preemptive, lower number = higher priority),
what is the waiting time for process P4?



9

4

0

8

12

Will be reviewed

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What is a semaphore?

A type of interrupt.

A hardware device for memory management.

A synchronization tool for controlling access to shared resources.

A file system structure.

✉ **Will be reviewed**

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What is a system call? 

An interface between a user program and the operating system kernel.

A memory allocation instruction.

A hardware interrupt.

A call to a user-defined function.

✉ **Will be reviewed**

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What is the purpose of the "mutex" semaphore in a typical Bounded-Buffer solution? 

To manage the producer process.

To count the number of items in the buffer.

To manage the consumer process.

To ensure mutual exclusion when accessing the buffer.

☑ **Will be reviewed**

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What is the Bounded-Buffer Problem? 🗉

A problem of coordinating producer and consumer processes sharing a fixed-size buffer.

A problem of managing disk I/O.

A problem of scheduling CPU processes.

A problem of managing memory allocation.

☑ **Will be reviewed**

18

What is the purpose of a process control block (PCB)? 🗉

To store information about a process, such as its state and program counter.

To store the executable code of a process.

To store the data used by a process.

To manage the file system.

☑ **Will be reviewed**

19

What is the main purpose of having different modes (User Mode and Kernel Mode) in an operating system? 

To provide security and prevent user programs from accessing critical system resources.

To increase the speed of user applications.

To simplify the operating system design.

To allow multiple users to run applications simultaneously.

 **Will be reviewed**

20

What is the purpose of the "mode bit" in the CPU? 

To control memory allocation.

To handle hardware interrupts.

To manage process scheduling.

To indicate whether the CPU is in User Mode or Kernel Mode.

 **Will be reviewed**

21

Consider the following processes with their arrival times and burst times:
Using Round-Robin (RR) scheduling with a time quantum of 2, what is the response time for process P2?



2

6

4

1

0

 **Will be reviewed**

22

What is "mutual exclusion" in the context of the critical-section problem? 

Preventing processes from entering the critical section.

Allowing processes to access any part of the memory.

Allowing multiple processes to access the critical section simultaneously.

Ensuring that only one process can execute its critical section at a time.

 **Will be reviewed**

23

Which scheduling algorithm can lead to "starvation" of low-priority processes?



Round Robin (RR)

Preemptive scheduling

First-Come, First-Served (FCFS)

 Priority Scheduling

 **Will be reviewed**

24

What event causes a process to transition from the "Running" state to the "Waiting" state? 

 I/O request.

 Process termination.

 Process creation.

 Time slice expiration.

 **Will be reviewed**

25

Which of the following is an example of a system call? 

 Performing a complex mathematical computation.

 Calculating the square root of a number.

 Printing text to the console.

 Sorting an array.

 **Will be reviewed**

26

What is the main purpose of Direct Memory Access (DMA)? 

To schedule processes.

To allow I/O devices to directly access system memory without CPU intervention.

To increase CPU processing speed.

To manage virtual memory.

 **Will be reviewed**

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Why does priority inversion occur? 

Due to inefficient CPU scheduling.

Due to the use of shared resources and synchronization mechanisms.

Due to incorrect memory allocation.

Due to hardware failures.



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